

Missing Cat Digital Forensics

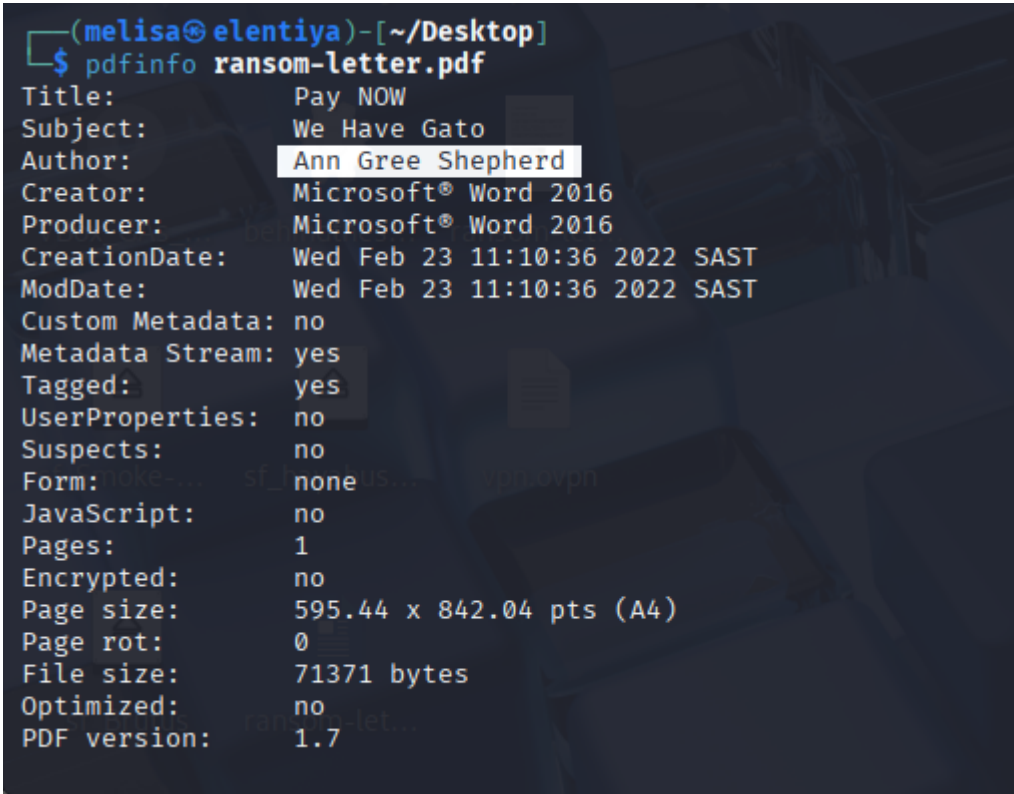
Scenario

My cat Gado has been abducted and the kidnapper has left a ransom note. The ransom note was in word form but it has been converted to pdf. In addition a Word image has been made.

Investigation

Document Metadata

An investigation was done to find out more about the kidnapper. Firstly, the metadata for the file was investigated.

A terminal window with a dark background and light blue text. The prompt is (melisa@elentiya) - [~/Desktop]. The command \$ pdfinfo ransom-letter.pdf has been entered. The output lists various PDF metadata fields and their values.

```
(melisa@elentiya) - [~/Desktop]
$ pdfinfo ransom-letter.pdf
Title:          Pay NOW
Subject:        We Have Gato
Author:         Ann Gree Shepherd
Creator:        Microsoft® Word 2016
Producer:       Microsoft® Word 2016
CreationDate:   Wed Feb 23 11:10:36 2022 SAST
ModDate:        Wed Feb 23 11:10:36 2022 SAST
Custom Metadata: no
Metadata Stream: yes
Tagged:         yes
UserProperties: no
Suspects:       no
Form:           none
JavaScript:     no
Pages:          1
Encrypted:      no
Page size:      595.44 x 842.04 pts (A4)
Page rot:       0
File size:      71371 bytes
Optimized:      no
PDF version:    1.7
```

Photo EXIF Data

Exchangeable Image File Format saves metadata to photos. It was relied on to find out information about what device was used to take the photo, the location the photo was taken and even camera settings. The tool used for this was: exiftool

```

(Metisa@etentiy) - [~/Desktop]
$ exiftool letter-image.jpg
ExifTool Version Number      : 13.25
File Name                    : letter-image.jpg
Directory                    : .
File Size                     : 127 kB
File Modification Date/Time   : 2022:02:23 10:53:33+02:00
File Access Date/Time        : 2025:09:07 22:31:46+02:00
File Inode Change Date/Time   : 2025:09:07 22:31:45+02:00
File Permissions              : -rwxr-xr-x
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Exif Byte Order               : Little-endian (Intel, II)
Compression                  : JPEG (old-style)
Make                         : Canon
Camera Model Name             : Canon EOS R6
Orientation                  : Horizontal (normal)
X Resolution                  : 300
Y Resolution                  : 300
Resolution Unit               : inches
Software                     : GIMP 2.10.28
Modify Date                   : 2022:02:15 17:23:40
Exposure Time                 : 1/200
F Number                     : 2.8
Exposure Program              : Manual
ISO                           : 640
Sensitivity Type              : Recommended Exposure Index
Recommended Exposure Index    : 640
Exif Version                  : 0231
Date/Time Original            : 2022:02:25 13:37:33

```

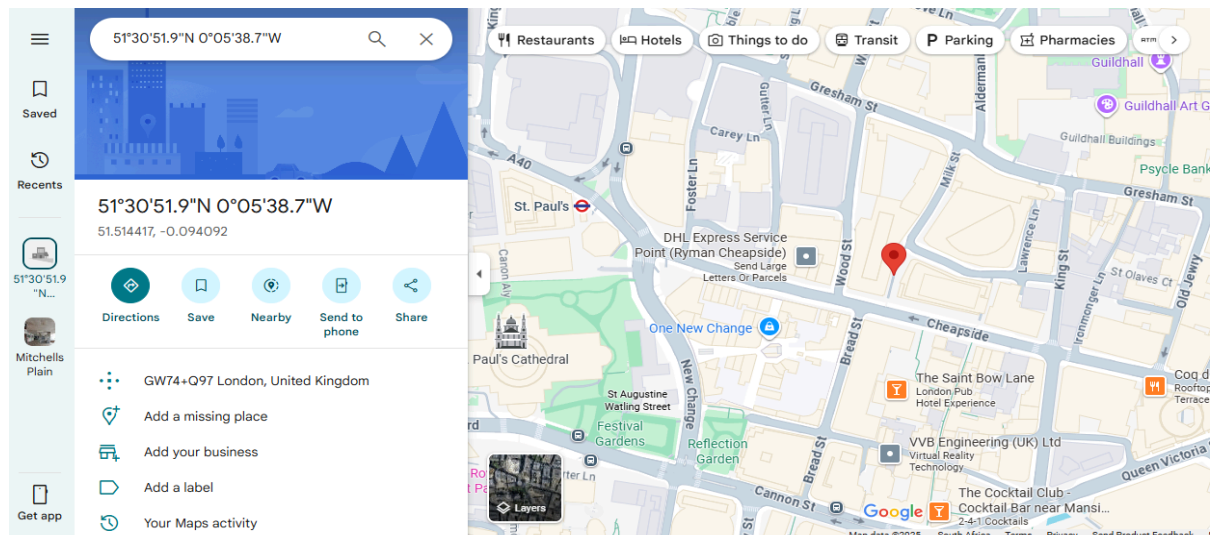
In addition, there was a parameter which showed the exact location the photo was taken.

```

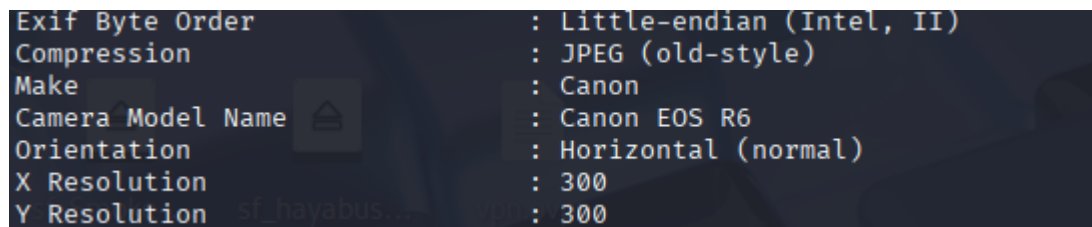
Thumbnail Image              : (Binary data 4941 bytes, use -b option to extract)
GPS Latitude                  : 51 deg 30' 51.90" N
GPS Longitude                 : 0 deg 5' 38.73" W
Date/Time Created            : 2022:02:15 17:23:40-17:23
Digital Creation Date/Time    : 2021:11:05 14:06:13+03:00
Circle Of Confusion          : 0.043 mm
Depth Of Field                : 0.06 m (0.76 - 0.82 m)
Field Of View                 : 54.9 deg
Focal Length                  : 50.0 mm (35 mm equivalent: 34.6 mm)
GPS Position                  : 51 deg 30' 51.90" N, 0 deg 5' 38.73" W
Hyperfocal Distance          : 20.58 m
Light Value                   : 7.9
Lens ID                       : Canon EF 50mm f/1.8 STM

```

These coordinates were inputted into Google Maps



In addition, the model name of the camera used was found:



Therefore the name of the kidnapper and her possible location could be found from the metadata of the ransom note and the ransom image.

Lessons Learned

- Pdftinfo tool is used to get pdf metadata which is sometimes preserved from word documents. This can reveal metadata information such as the time of creation and the author of the document.
- Exiftool is a metadata tool which extracts information for various file formats. In this case it contained details of the camera used to take the picture and the location of the photo was taken
- Metadata is very important and is used in digital forensics as one of the information sources.